

Ayush Singh

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WORK EXPERIENCE

– ITax Easy Pvt Ltd

Full Stack Developer Intern

March 2025 – May 2025

- Designed and optimized RESTful APIs, reducing average response time by **30%+** through efficient query handling and request processing.
- Implemented secure authentication workflows using JWT, improving session management and system security.
- Integrated CI/CD pipelines using GitHub Actions, enhancing deployment reliability and reducing release inconsistencies by **15%+**.
- Diagnosed and resolved production-level issues, improving application stability and frontend-backend interaction performance.

– Cloud Computing Cell

Backend Developer

Oct 2024 – Present

- Developed and optimized RESTful APIs in a collaborative 45-member team, improving system performance by **45%+** through efficient database queries and backend optimizations.
- Designed and refined database workflows using optimized schema design and query tuning to reduce latency and improve data retrieval efficiency.
- Improved system reliability by debugging production-level issues and implementing structured error handling and performance optimizations.
- Collaborated on modular backend architecture, enabling scalable feature development and maintainable codebase design.

PROJECTS

– IMAGO [\[Live-Link\]](#) Spring Boot, MongoDB, REST APIs, JWT, Docker, ML Integration

[Source Code](#)

Designed and developed a scalable Spring Boot REST API for image upload and transformation workflows. Implemented ML-based background removal pipeline to automate image processing tasks. Built optimized image operations including compression, resizing, cropping, and format conversion. Secured endpoints using JWT-based authentication. Designed efficient MongoDB schema with indexing to optimize query performance. Containerized the backend using Docker for consistent and scalable deployment.

– Smart Airport Ride Pooling System Spring Boot, PostgreSQL, Redis, Docker, JPA, Flyway

[Source Code](#)

Designed a concurrency-safe backend system to group passengers into shared rides using geo-based matching algorithms. Implemented Redis-based distributed locking to prevent duplicate ride assignments under concurrent requests. Applied optimistic locking (@Version) to ensure seat consistency and prevent overbooking. Built multi-stage matching engine using geo-filtering, eligibility checks, and detour-based scoring. Designed dynamic pricing engine using Strategy Pattern (base fare, distance, surge pricing, ride-sharing discount). Ensured transactional safety and high concurrency handling with zero race conditions. Containerized services using Docker and managed schema migrations via Flyway.

– RepoScribe Spring Boot, MongoDB, OAuth2, REST APIs, Docker, NLP Integration

[Source Code](#)

Developed secure OAuth2-based GitHub authentication flow to fetch and process repository metadata. Built automated documentation pipeline using NLP APIs for generating context-aware summaries. Designed MongoDB aggregation pipelines for efficient large-scale repository data processing. Implemented asynchronous processing concepts for handling repository analysis workflows. Containerized application using Docker to ensure consistent deployment across environments.

TECHNICAL SKILLS

Languages:Java, JavaScript

Backend:Spring Boot, REST API Design, Microservices Architecture

Frontend:ReactJS

Databases:MongoDB, PostgreSQL, MySQL

DevOps & Tools:Docker, GitHub Actions

Monitoring & Observability:Prometheus, Grafana

Cloud:AWS (EC2, S3)

Core Concepts:System Design (High Level), Caching (Redis), Authentication (JWT, OAuth2), Rate Limiting

EDUCATION

Ajay Kumar Garg Engineering College (AKGEC), Ghaziabad, Uttar Pradesh, India

Oct 2023 – Oct 2027

Bachelor of Technology in Computer Science (CGPA: 8.2)

ACHIEVEMENTS

- Ranked in the **Top 100** among **1000+** teams at **Executive 4.0 Hackathon (DTU)** by building a scalable backend solution
- Solved **500+** **DSA** problems on **LeetCode** and **CodeChef**